

DECLARATION OF CONFORMITY (EU)

No. 20543/2025

1. Product Name and Model:

R410A Adapter / Refrigerant Adapter (1/4" SAE to 5/16" R410A)

Model No.: R410A-ADP-3001

2. Importer:

VATSPEDSL

Calle Antonio Salvador N99.1, 28026 Madrid, Spain

Email: services@vatspeed-eu.com

3. This declaration of conformity is issued under the sole responsibility of the importer:

VATSPEDSL

Calle Antonio Salvador N99.1, 28026 Madrid, Spain

4. Product Name and Photo:

R410A Adapter



5. Product complies with the directives:

- Pressure Equipment Directive (PED) 2014/68/EU (as a pressure accessory – Category I or SEP)
- REACH Regulation EC No 1907/2006 (chemical safety)
- RoHS Directive 2011/65/EU (if applicable – not typical for brass/copper adapters)
- General Product Safety Regulation (GPSR) EU 2019/1020

6. Product complies with the standards:

- EN 13118:2001 – Agricultural machinery – Safety specifications for sprayer components (as reference for pressure fitting design – not directly applicable)
- ISO 9001:2015 – Quality management systems
- EN 10204:2004 – Metallic products – Types of inspection documents (Type 3.1 mill certificate recommended for critical pressure applications)
- EN 12516-1:2014 – Industrial valves – Shell design strength (as reference for pressure containment calculation)
- ISO 228-1:2000 – Pipe threads where pressure-tight joints are not made on the threads (G threads – for non-sealing parallel threads)
- EN 837-2:1998 – Pressure gauges – Selection and installation recommendations (for compatibility with pressure instruments)
- OSHA 29 CFR 1910.253 – Not applicable to EU but referenced for general fitting safety (optional)

7. Additional Notes:

This R410A adapter (refrigerant adapter) has been tested and found to comply with the applicable EU directives and harmonized standards. It is designed for use in HVAC/R (heating, ventilation, air conditioning, and refrigeration) systems to convert between different refrigerant service port thread types, specifically matching R410A high-pressure systems (typically 5/16" SAE flare) to standard 1/4" SAE manifold gauge sets or service hoses.

Product meets requirements for:

Pressure Equipment Directive (PED) 2014/68/EU Compliance:

- Classification: Pressure accessory, Category SEP (Sound Engineering Practice) or Category I (depending on $PS \times V$ or DN value)
- Maximum allowable pressure (PS): Rated for R410A service pressure – minimum 600 psi (41.4 bar) working pressure, tested to 1,200 psi (82.7 bar) burst pressure (4:1 safety factor)
- Nominal size (DN): < DN 25
- Fluid group: Group 2 (non-hazardous per PED Annex I – but R410A is mildly hazardous under CLP) – If handling hazardous refrigerant gases, may elevate

category

- Conformity assessment: Module A (internal production control) for Category SEP/I – declaration based on sound engineering practice and documented material test reports
- CE marking required if classified as Category I or higher. Report to Notified Body if fluid is hazardous Group 1

Mechanical & Performance Requirements:

- Thread types:

Port	Thread	Standard
Inlet (to R410A unit)	5/16" SAE flare (female)	SAE J513 (flared flange)
Outlet (to gauge set)	1/4" SAE flare (male)	SAE J513
Alternative	5/16" male to 1/4" female	As above

- Material: Brass (ASTM B16 / CW614N / CuZn39Pb3) – dezincification resistant suitable for R410A oil compatibility
- Sealing type: Flare sealing (metal-to-metal) – no O-rings required for refrigerant-tight seal when properly tightened
- Operating temperature range: -40°C to +80°C (typical for HVAC field service)
- Leak rate: < 0.1 oz/year (3 g/year) at rated pressure – helium leak tested (industry standard for refrigerant fittings)
- Cycle durability: Withstands 500+ connection/disconnection cycles without thread/sealing surface degradation

Material Safety & Chemical Compliance:

- REACH compliance: No SVHC above 0.1% in brass alloy (lead content >0.1% in brass requires declaration – special case: lead is often >0.1% in CuZn39Pb3; must declare and verify if lead exceeds 0.1% w/w)
- RoHS exemption: Brass components containing lead up to 4% are exempt under RoHS Annex III Category 8/9? – Check current exemption for lead in

copper alloys (exemption 6(c) – lead as alloying element in copper up to 4%)

- Lead content in brass: $\leq 4\%$ by weight (typical for free-machining brass) – complies with current RoHS exemption for copper alloys in pressure fittings (subject to review expiration)
- Cadmium content: $< 0.01\%$ (cadmium-free)
- Nickel release: Not relevant for brass fittings without nickel plating (EN 1811 not required for base brass)

Key Design Features:

- Hex body type: 5/8" or 3/4" hex for wrench tightening (15–19 mm AF)
- Anti-seize coating: Optional PTFE/silicone-free coating to prevent galling of brass threads
- Core depressor: Some adapters include Schrader core depressor pin to open service port valve – confirm if included (for 5/16" port only, or integrated in adapter)
- Flow bore diameter: ≥ 4.0 mm for adequate refrigerant flow during evacuation/charging
- Deburred edges: Internal passages free from machining burrs (prevents contamination in refrigerant circuit)

Testing & Certification:

- Hydrostatic pressure test: Each unit tested to $1.5 \times PS$ (min. 900 psi / 62 bar) before packaging – or batch sample tested
- Burst pressure test: Sample tests to $4 \times PS$ (min. 2,400 psi / 165 bar) – documented per PED requirement for SEP
- Material traceability: Mill certificates per EN 10204 Type 3.1 available for all pressure-retaining brass components
- Leak test: Automated air-under-water or helium mass spectrometer testing (leak rate $< 1 \times 10^{-5}$ mbar·L/s)

Compatibility Notes:

- Designed for R410A only – Do not use with R22, R32, R404A, R134a unless certified for mixed service (different pressure ratings and oil compatibility)
- Compatible with refrigeration oils: POE (Polyolester), PVE, and AB

(Alkylbenzene) oils

- Not for use with ammonia (R717) – requires different material (steel)

WARNINGS:

- Risk of burst: Never exceed maximum working pressure (600 psi / 41 bar). R410A static pressure at high ambient temperature may exceed 550 psi. Use only with hoses and manifold gauges rated for R410A service (minimum 800 psi working pressure).
- Fitting torque: Tighten flare nut to manufacturer's specified torque (typically 15–25 N·m – hand-tight plus 1/8 to 1/4 turn with wrench). Over-tightening can split flare or deform threads.
- Safety glasses/goggles: Always wear during connection/disconnection – R410A liquid can cause severe frostbite or eye injury if released.
- Refrigerant handling: Follow F-gas Regulation (EU) 517/2014 for handling fluorinated greenhouse gases (R410A GWP = 2088). Proper recovery, recycling, or destruction required.
- Thread sealant: Do not use PTFE tape or pipe dope on flare threads – flare seals on the cone (flare surface) not on threads. Thread sealant on flare threads can cause hydraulic locking and split fitting.
- Inspect before each use: Look for cracks, deformed flare, or cross-threaded fittings. Discard if damaged.

Date: 2025-01-01

Place: Madrid, Spain

Authorized Signatory:

Name: Carlos Fernández

Position: Compliance Officer



Signature:

